

Local Search: of Hill Climbing With random Walk & Random Restart

Part-5

Introduction

Local search algorithms offer an intuitive approach to optimization problems, often mimicking how one might explore a landscape to find the highest peak. However, this intuitive nature also presents significant challenges. A primary issue is the tendency for these algorithms to get trapped in "local optima"—solutions that are better than their immediate neighbors but not the best overall solution in the entire search space. This lecture delves into various advanced techniques designed to overcome these limitations, transforming basic hill climbing into more robust and effective search strategies. By introducing elements of memory, systematic exploration, and controlled randomness, these methods aim to enhance the completeness and efficiency of local search, making them powerful tools for tackling complex problems in artificial intelligence and beyond. Understanding these variations is crucial for anyone looking to apply optimization techniques effectively, especially when faced with time-constrained scenarios or highly rugged search landscapes.

Core Concepts

Term	Definition	Significance
Tabu Search	A metaheuristic that enhances local search by maintaining a fixed-length queue of recently visited states (the "Tabu list") to prevent the algorithm from immediately revisiting them, thereby avoiding cycles and encouraging exploration.	Prevents cycling between states of equal value and forces the search to explore new regions, improving the chances of escaping local optima.
Enforced Hill Climbing	An algorithm that combines hill climbing with breadth-first search (BFS). When stuck in a local optimum, it performs BFS until it finds a state better than	Offers a middle ground between purely local and systematic search, ensuring progress by systematically escaping

Term	Definition	Significance
	the point where hill climbing began, then jumps to that state and restarts hill climbing.	local optima, albeit with higher memory and time costs during BFS phases.
Asymptotically Complete	Refers to an algorithm that is guaranteed to find an optimal solution given enough time, even if that time is theoretically infinite.	Contrasts with greedy local search, which is not asymptotically complete as it can get permanently stuck in local optima. This property is a goal for more robust search methods.
Random Walk Hill Climbing	A stochastic local search method where, at each step, the algorithm moves to a greedy neighbor with a certain probability (P) and to a random neighbor with the complementary probability ($1-P$).	Introduces randomness to help escape local optima by occasionally taking suboptimal moves, balancing exploitation (greedy) with exploration (random).
Random Restart Hill Climbing	An algorithm that repeatedly performs hill climbing from different, randomly sampled starting states, always keeping track of the best solution found across all restarts.	A simple yet powerful technique to mitigate the problem of local optima by increasing the chance of starting in a basin of attraction that leads to a better global optimum.
Simulated Annealing	A metaheuristic inspired by the annealing process in metallurgy, which uses a slowly changing probability schedule for accepting suboptimal moves, allowing for more exploration early on and more exploitation later.	Provides a more sophisticated way to manage the trade-off between exploration and exploitation, offering a theoretically sound approach to finding near-optimal solutions.
Local Optimum		

Term	Definition	Significance
	A state in the search space that is better than all its immediate neighbors, but not necessarily the best state in the entire search space (global optimum).	The primary challenge for basic hill climbing algorithms, as they tend to get stuck here, necessitating more advanced techniques to escape.

Detailed Analysis

1. Overcoming Cycles and Sideways Moves

Basic hill climbing can suffer from non-termination, particularly when encountering "sideways moves" – transitions between states of equal value. This can lead to infinite loops, such as oscillating between two states (A-B-A-B) or getting trapped in a plateau of equally good states. To address this, a simple fix is to impose a limit on the maximum number of sideways moves allowed.

A more sophisticated approach is **Tabu Search**. This algorithm introduces a "Tabu list," which is a fixed-length queue storing recently visited states. The search is then prohibited from revisiting any state currently on this list. For example, if the last 100 visited nodes are on the Tabu list, the algorithm cannot move to any of those nodes. This mechanism prevents the algorithm from cycling and encourages it to explore new parts of the search space. While a very large Tabu list would effectively turn the algorithm into a systematic search (by preventing all redundancy), in practice, a small, constant-sized Tabu list (e.g., 100 states) is used to balance exploration with efficiency. The most recent state is added to the queue, and the oldest is dropped, maintaining the fixed length.

2. Enforced Hill Climbing: A Hybrid Approach

When hill climbing gets stuck in a local optimum, **Enforced Hill Climbing** provides a mechanism to escape. The core idea is that upon reaching a local optimum, the algorithm switches from local hill climbing to a breadth-first search (BFS). This BFS explores the search space systematically in all directions from the local optimum. The BFS continues until it discovers a state whose value is better than the initial state from which the current hill climbing phase began. Once such a "better" state is found, the algorithm "jumps" to it, discards the BFS memory, and restarts hill climbing from this new, improved position.

This approach enforces progress by ensuring that the algorithm will eventually find a path out of any local optimum, provided the search space is connected. In practice, the BFS phase can be computationally intensive, consuming significant memory and time. This makes Enforced Hill Climbing a middle ground between purely local search (which has minimal memory requirements but can get stuck) and systematic search (which guarantees completeness but has high memory and time costs). A notable application of this algorithm was in the FF planner, a highly successful AI planning system in the early 2000s, which used it to navigate complex path spaces by making local moves (e.g., removing and adding edges) and escaping local optima.

3. The Need for Stochasticity: Beyond Deterministic Search

Deterministic local search algorithms, like basic hill climbing, are not "asymptotically complete"—they are not guaranteed to find the optimal solution even with infinite time, as they can get permanently stuck in local optima. In contrast, algorithms like random sampling (randomly picking new states) and random walk (randomly moving to a neighbor without considering objective function) are asymptotically complete, meaning they will eventually find the optimum if the search space is connected. However, these purely random methods lack the efficiency and directedness of greedy hill climbing.

This highlights a crucial trade-off: efficiency versus completeness. The goal then becomes to design hybrid algorithms that combine the directedness of greedy search with the completeness offered by randomness, aiming to avoid local optima while still making progress.

4. Stochastic Hill Climbing: Introducing Randomness

To create algorithms that are both efficient and asymptotically complete, randomness is introduced into hill climbing. These are broadly termed stochastic hill climbing algorithms.

- **Random Walk Hill Climbing:** This variation modifies the decision-making at each step. Instead of always taking the greedy move, the algorithm takes a greedy step with a certain probability P and a random step (moving to any random neighbor) with probability $1-P$. This allows the algorithm to occasionally make suboptimal moves, which can be crucial for escaping local optima. Ideally, the probability P should change over time; it should start lower (allowing more exploration to escape bad initial hills) and gradually increase (favoring exploitation as the search progresses).

- **Random Restart Hill Climbing:** This is often considered one of the most practical and effective local search algorithms. The strategy is straightforward: perform hill climbing until a local optimum is reached. Then, instead of trying to escape from that specific local optimum, the algorithm randomly samples an entirely new starting state and begins hill climbing again. This process is repeated multiple times, always keeping track of the best solution found across all restarts. The underlying principle is "if at first you don't succeed, try, try again." The number of restarts can be fixed or determined by a time limit. If each hill climbing run has a probability P of finding a satisfactory solution, the expected number of restarts needed to find a solution is $1/P$. This makes it a robust choice for many problems, especially when given a fixed time budget.
- **Hybrid Approaches:** It's also possible to combine these ideas, for example, by performing hill climbing with both random walk and random restarts. At each step, the algorithm might choose between a greedy move, a random walk move to a neighbor, or a complete random restart to a new state, each with its own probability.

5. Simulated Annealing: A Refined Stochastic Approach

Simulated Annealing is a more advanced stochastic hill climbing algorithm that refines the idea of changing the probability of taking random steps over time. Inspired by the annealing process in metallurgy (where a material is heated and then slowly cooled to reduce defects), Simulated Annealing starts with a high probability of accepting suboptimal moves (high "temperature"), allowing for broad exploration of the search space. As the search progresses, this probability gradually decreases (the "temperature" cools), making the algorithm more likely to accept only improving moves and converge towards a local optimum. This controlled cooling schedule provides a powerful mechanism to balance exploration and exploitation, making it effective for finding high-quality solutions in complex landscapes.

Key Takeaways

- [] Basic hill climbing can get stuck in local optima and suffer from non-termination due to sideways moves.
- [] Tabu Search uses a fixed-length memory (Tabu list) to prevent revisiting recently explored states, thereby avoiding cycles and encouraging broader exploration.

- [] Enforced Hill Climbing combines local search with systematic breadth-first search to escape local optima, representing a middle ground between purely local and systematic methods.
- [] Deterministic local search algorithms are not asymptotically complete, while purely random methods (random walk, random sampling) are, but lack efficiency.
- [] Stochastic hill climbing algorithms introduce randomness to achieve both efficiency and asymptotic completeness.
- [] Random Walk Hill Climbing balances greedy moves with random moves, with the probability of greedy moves ideally increasing over time.
- [] Random Restart Hill Climbing is a highly practical strategy that repeatedly runs hill climbing from new random starting points, keeping the best solution found.
- [] The expected number of restarts needed for Random Restart Hill Climbing is $1/P$, where P is the probability of success for a single run.
- [] Simulated Annealing is a sophisticated stochastic method that uses a slowly changing probability schedule to accept suboptimal moves, balancing exploration and exploitation.

Conclusion

The evolution of local search from basic hill climbing to sophisticated stochastic methods like Tabu Search, Enforced Hill Climbing, Random Walk, and Random Restart demonstrates a continuous effort to overcome inherent limitations and enhance performance. These algorithms provide practical solutions for navigating complex, multi-modal search landscapes where global optima are elusive. By strategically incorporating memory, systematic exploration, and controlled randomness, they offer a robust toolkit for optimization. The success of algorithms like Enforced Hill Climbing in competitive AI planning systems, such as the FF planner, underscores their real-world impact. Ultimately, the choice of algorithm often depends on the problem's specific constraints, particularly time limits. In scenarios where a quick, high-quality solution is needed within a fixed timeframe, these adaptive local search techniques, especially Random Restart Hill Climbing, prove invaluable, allowing practitioners to effectively utilize available computational resources to find the best possible outcome. The journey continues with even more advanced techniques like Simulated Annealing and genetic algorithms, which further refine the balance between exploration and exploitation.